Hall Ticket Number:

MR-22

Code: B325CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Object Oriented Programming through Java

(Common to CSE, CSB and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1= 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	"JVM" is platform dependant. Justify.	1	1	1	1
1. b)	Explain the role of java garbage collection in memory management.	1	2	1	1,2
1. c)	Why is inheritance important?	1	2	2	1,2
1. d)	Explain private and protected keywords.	1	2	2	1
1. e)	List the methods in Thread class.	1	1	3	1
1. f)	What is the difference between error and exception?	1	1	3	1,2
1. g)	What is an adapter class?	1	1	4	1,2
1. h)	Name three sub classes of component class.	1	3	4	1
1. i)	Define applet.	1	2	5	1
1. j)	What is an event in java AWT?	1	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the features of Java in detail.	5	2	1	1,2,3
2. b)	How can you find the maximum and minimum values in an array in Java?	5	1	1	1,2,3
	OR				
2. c)	What are the drawbacks of procedural languages? Explain the need of object oriented programming with suitable program.	5	3	1	1,2,3
2. d)	Define overloading. How to overload constructor? Explain in detail.	5	2	1	1,2,3
3. a)	Is multiple inheritance allowed in Java? Justify your answer with suitable example.	5	3	2	1,2,3
3. b)	Write in detail about creating and accessing packages.	5	2	2	1,2,3
	OR				
3. c)	Differentiate between abstract class and interfaces.	5	3	2	1,2,3
3. d)	Explain Runtime polymorphism with the help of suitable example.	5	3	2	1,2,3
4. a)	Explain the following clauses i)try ii)throw iii)catch.	5	2	3	1,2,3
4. b)	Explain thread life cycle with a neat labelled diagram.	5	2	3	1,2,3
	OR				
4. c)	Explain thread synchronization with the help of suitable program.	5	2	3	1,2,3
4. d)	Write a Java program to implement Runnable class to create a thread.	5	6	3	1,2,3
5. a)	Explain Delegation Event model with the help of an example.	5	3	4	1,2,3
5. b)	Write a program to demonstrate Mouse Events.	5	3	4	1,2,3
	OR				
5. c)	Explain the following AWT components i)TextField ii)TextArea iii)Label	5	2	4	1,2,3
5. d)	Explain BorderLayout Manager with the help of suitable program.	5	3	4	1,2,3

P.T.O.

6. a)	Explain about JTree with the help of suitable example.	5	3	5	1,2,3
6. b)	Explain how parameter is passed to Applet.	5	2	5	1,2,3
	OR				
6. c)	Explain in detail about classification of swing components.	5	2	5	1,2,3
6. d)	Differentiate between AWT and swings.	5	2	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome